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2S-TOOL: a new open-access tool for the automatic analysis of stress–strain curves for aquifer system characterization

María I. Navarro-Hernández¹ · Sergio García-Pozo^{1,2} · Javier Valdes-Abellan¹ · Roberto Tomás¹

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Abstract

Groundwater level variations induce aquifer–system deformation, either elastic or inelastic, which can be assessed through stress–strain curves. These curves, which represent piezometric level variations against ground deformation, provide key insights into the mechanical behaviour of aquifer systems and allow the estimation of storage coefficients. Traditionally, their interpretation relies on visual assessment, making it subjective and time-consuming. This work presents 2 S-TOOL, an open-access MATLAB® application that automates stress–strain curve analysis and storage coefficient estimation, reducing manual effort and bias. The tool integrates groundwater level data from observation wells and ground deformation measurements from in situ or remote sensing techniques, automatically identifying elastic and inelastic segments and calculating the S_{ke} and S_{kv} coefficients. 2 S-TOOL has been tested via real-world case studies, which demonstrate its ability to streamline analysis and improve consistency. The software provides a reproducible, efficient, and objective method for stress–strain curve interpretation, enhancing hydrogeological modelling and groundwater management applications.

Keywords Aquifer-system deformation · Stress–strain curve · Groundwater extraction · Storage coefficient · Land subsidence · MATLAB

Introduction

Land subsidence induced by intensive groundwater withdrawal from detrital aquifers is a worldwide issue affecting many important cities, such as Jakarta, Beijing, Mexico City, Venice, etc., where the aquifer exploitation has led to

critical subsidence issues (Carbognin et al. 2004; Chen et al. 2019; Harintaka et al. 2024; Zapata-Norberto et al. 2025). The expanding urban population exacerbates this issue, thereby intensifying the strain on already limited and occasionally scarce freshwater resources (Galloway et al. 1999; Galloway and Burbey 2011; Herrera-García et al. 2021).

The subsidence and uplift of aquifer systems can be measured via in situ instrumentation, with borehole extensometers and levelling survey techniques commonly employed (Galloway and Burbey 2011; Tomás et al. 2014a, b). Precise levelling typically achieves a vertical accuracy between 0.1 and 1.0 mm, following a line-network configuration, with measurements generally taken on a monthly or annual basis. Borehole extensometers offer a much higher precision, in the order of 0.01 to 0.1 mm in the vertical component, measured at a single point, and can provide continuous or daily records. Continuous GNSS stations have a precision of approximately 1–2 mm for both vertical and horizontal components, usually operating as part of a network or at a single location, with continuous data acquisition. In contrast, GNSS benchmarks, which are measured less frequently (monthly or annually), have a lower

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✉ María I. Navarro-Hernández
mainnahe@ua.es

Sergio García-Pozo
sgp107@alu.ua.es; sergio.gp@p14cimentaciones.es

Javier Valdes-Abellan
javier.valdes@ua.es

Roberto Tomás
roberto.tomas@ua.es

¹ Department of Civil Engineering, Polytechnic School, University of Alicante, Alicante, Spain

² P14 Cimentaciones, University of Alicante, C/ Campanar, nº 6 Bajo 46920 Mislata, Valencia, Spain

precision of around 8–15 mm, but also capture both vertical and horizontal displacements (Galloway et al. 1998). In recent decades, interferometric synthetic aperture radar (InSAR) has become a widely used technique for measuring ground deformation in aquifer systems (Raspini et al. 2022). This technique can detect range-direction displacements with sub-millimetric precision (typically < 1 mm) at the pixel scale of a map, with a temporal resolution ranging from weekly to monthly, depending on satellite acquisition schedules (Navarro-Hernández et al. 2022; Tomás et al. 2014a). Ground levels, on the other hand, are measured via piezometers. However, the availability of precise displacement and groundwater-level measurements does not, by itself, guarantee that observed ground displacements are related to aquifer-system compaction.

It is therefore important to highlight that, to ensure measured displacements are linked to aquifer-system compaction, the displacement time series are usually cross-analysed or compared with groundwater level fluctuations. Complementarily, deformation signals related to other contributing factors such as tectonic, construction, or seasonal effects can be removed from the general deformation trends using spatial filtering or ancillary datasets (Amelung et al. 1999; Song et al. 2022).

The representation of the deformations of the aquifer system on the ordinate axis and the groundwater level on the abscissa axis provides the stress–strain curves of the aquifers. The use of stress–strain curves is a graphical way to describe the relationship between ground displacement due to aquifer–system compaction and groundwater level evolution over time (Riley 1969). The most important characteristics of the mechanical behaviour of an aquifer system or of a specific layer of the aquifer system can be interpreted through the information contained in the stress–strain curves. Furthermore, the values of some hydrogeological variables, such as the storage coefficient, can be estimated from this relationship. Nevertheless, despite the advantages and information offered by the stress–strain relationship, the analysis of these curves is typically conducted manually without a clear and objective protocol (Burbey 2001; Ezquerro 2021; Galloway et al. 1999). The process of manual analysis is tedious and time-consuming and leads to subjective results highly reliant on the researcher's ability and expertise to interpret graphical information.

Although various tools have been developed in recent years for the analysis and modelling of hydrogeological systems, none of them, to our knowledge, offer a fully automated procedure for generating and interpreting stress–strain curves from time series of groundwater levels and ground displacements, with the specific aim of estimating the elastic and inelastic skeletal storage coefficients. While the general concept of stress–strain curve analysis may

appear straightforward, in practice it involves several technical challenges, such as managing asynchronous datasets, handling missing or noisy measurements, and objectively identifying elastic loops within complex deformation patterns. These steps are often carried out manually, making the process time-consuming and prone to subjectivity. To avoid these operational issues, in this work we propose an open-source MATLAB-based application (2 S-TOOL), which automates both the construction of these curves from time series and the parameter estimation process through an algorithm capable of reliably distinguishing between elastic and inelastic segments. This ensures a more consistent and reproducible analysis, reduces interpretative bias, and simplifies a methodology that has traditionally depended on expert visual inspection. The application has been validated using real data from aquifer systems affected by land subsidence worldwide.

Stress–strain curves

The stress–strain curves of aquifer systems represent the relationship between ground displacement measurements (land subsidence in this case) and piezometric level fluctuations during a common observation period. This method was developed by Riley (Riley 1969), who proposed plotting the stress (i.e., groundwater level variations) against the strain (i.e., displacements) of an aquifer system or a specific layer of an aquifer to estimate the specific skeletal storage coefficient of compressible strata (Burbey 2001).

A key aspect for interpreting these curves lies in identifying the threshold beyond which deformations become irreversible, an analysis that begins with understanding the preconsolidation pressure. This concept refers to the maximum effective stress ($\sigma_{e(\max)}$) experienced by a soil throughout its geological history. This pressure distinguishes between elastic (recoverable) and inelastic (irreversible) deformations of the soil. This concept is fundamental for understanding the mechanisms of land subsidence caused by fluctuations in the piezometric level (Poland 1984). In soils affected by groundwater withdrawal, the preconsolidation pressure indicates the maximum stress induced by a decrease in the piezometric level. This value suggests the groundwater depth or elevation at which deformations become irreversible. In other words, once the preconsolidation pressure is reached, a portion of the displacement cannot be recovered (Hoffmann 2003; Tomás et al. 2007).

In 1925, Terzaghi defined the poroelasticity theory, which is related to the water level fluctuation (hydraulic head) and the effective stress in the aquifer skeleton (Galloway and Hoffmann 2007; Hoffmann et al. 2001) as follows:

$$\Delta \sigma_e = \Delta \sigma_T - \rho \Delta h \quad (1)$$

where $\Delta \sigma_e$ is the effective stress change ($\text{ML}^{-1}\text{T}^{-2}$), σ_T is the total stress change ($\text{ML}^{-1}\text{T}^{-2}$), ρ is the water density ($\text{ML}^{-2}\text{T}^{-2}$) and Δh is the piezometric level (fluid pressure) change (L). Furthermore, it should be noted that the units used for the different variables must be consistent. From this equation, when the total stress remains constant ($\Delta \sigma_T = 0$), changes in the piezometric level are equivalent to changes in the effective stress, which can result in either compression or expansion of the aquifer skeleton, depending on the new load (Galloway et al. 1998). When the effective stress decreases due to an increase in the fluid pressure (i.e., the piezometric level increases), the aquifer expands elastically. When the effective stress increases because the fluid pressure decreases but the effective stress does not exceed the maximum past value, the aquifer system compresses elastically. However, if the maximum past effective stress is exceeded, then the deformation is inelastic, and the reduction in pore volume results in compaction (Galloway et al. 1998; Hoffmann et al. 2001).

In the hydrogeologic characterization of aquifers, skeletal specific storage is crucial in compressibility estimation. The compressibility can be elastic or inelastic for aquifers and aquitards in an aquifer system and can be expressed in skeletal specific storage (S_{sk}) terms (Galloway et al. 1998):

$$\begin{cases} S_{sk} = S_{ske} = \alpha_{ke} \rho g & \text{when } \sigma_e < \sigma_{e(\max)} \\ S_{sk} = S_{skv} = \alpha_{kv} \rho g & \text{when } \sigma_e > \sigma_{e(\max)} \end{cases} \quad (2)$$

where the subscripts e and v correspond to elastic and virgin (inelastic), respectively; ρ is the fluid density; and g is the acceleration of gravity. The storage coefficient (S) is the volume of water released from the aquifer system per unit surface area when there is a unit change in piezometric level (L^3/L^3), and the specific storage coefficient (S_s) is the amount of water stored or released per unit surface area per unit change in piezometric level per unit thickness (L^{-1}) (Galloway and Hoffmann 2007). Furthermore, under confined conditions, the specific storage is composed of two terms, as shown in Eq. [3], where S_{sk} is a component due to compressibility of the skeletal unit and where S_{sw} is a component due to the compressibility of the water (Riley 1969). Considering that S_s' corresponds to the compressibility of the aquitard layers, S_s corresponds to the compressibility of the aquifer layers, and their sum represents the aquifer system skeleton S_{sk} .

$$S_s = S_s' + S_s + S_w = S_{sk} + S_{sw} \quad (3)$$

However, for compact aquifer systems $S_{sk} \gg S_w$, the changes in storage caused by changes in the piezometric level are mainly determined by skeletal compressibility

(Galloway and Hoffmann 2007). Thus, Eq. [3] can be rewritten as:

$$S_s = S_{sk} \quad (4)$$

Considering the product of the elastic or inelastic skeletal specific storage and aggregate thickness of the aquitards or aquifers $\sum b$, where b is the thickness of a specific layer, the skeletal storage coefficient S_k can be defined as (Galloway et al. 1998):

$$\begin{cases} S_k = S_{ke} = S_{ske}(\sum b) & \text{when } \sigma_e < \sigma_{e(\max)} \\ S_k = S_{kv} = S_{skv}(\sum b) & \text{when } \sigma_e > \sigma_{e(\max)} \end{cases} \quad (5)$$

Riley (1969) demonstrated that when the effective stress changes gradually over time while the total stress remains constant, S_k can be calculated from paired time series of piezometric changes and displacements (Galloway and Hoffmann 2007) by:

$$S_k = \frac{\Delta b}{\Delta h} \quad (6)$$

In this formulation, S_k represents a vertically averaged storage coefficient for the entire thickness of the compacting unit, as determined from the aggregate thickness ($\sum b$). This simplification reflects the limitations of most available field datasets, which rarely provide sufficient depth-resolved deformation measurements to allow for accurate vertical discretization. Where data permit, the method can be applied separately to individual depth intervals to capture stratified mechanical behaviour.

The formulation presented in Eqs. [2]–[6] follows the classical confined-aquifer framework, where skeletal specific storage reflects saturated zone compressibility. For unconfined aquifers, the method can be applied if subsidence is dominated by saturated-zone compaction, in which case the contribution of the unsaturated zone compressibility is assumed negligible. In these cases, specific yield does not explicitly appear in the equations, as storage parameters are derived directly from the deformation–piezometric level relationship.

Mechanical behaviour of aquifers

Considering the different variables participating in the phenomenon, such as the stratigraphic features of the aquifer system, the pumping conditions, the climate setup, particularly the precipitation trend and the past groundwater levels, which define the preconsolidation head, aquifers can exhibit different mechanical behaviours, and even more, the same aquifer can exhibit different behaviours at different

periods of time. From the analysis of previous studies on the relationship between subsidence and groundwater level evolution via stress–strain curves, four main behaviour types can be identified: elastic, plastic, elastoplastic and viscoelastic–plastic.

When the stress state in a layer of an aquifer system does not exceed the preconsolidation stress (i.e., the previous maximum effective stress that has been sustained in the past) or when the water level is always above its minimum historical level, the system exhibits fully recoverable elastic deformation. This is the typical case of aquifer systems exhibiting seasonal variations in groundwater level, with an annual periodicity and normally without an important effect caused by human exploitation. Deformation is associated with both compaction and expansion, resulting in curves that exhibit synchronized or consistently delayed deformation in response to groundwater level fluctuations. However, in natural environments, aquifer systems do not often exhibit perfect elastic behaviour since there is usually a small plastic nonrecoverable deformation component when two consecutive and theoretically elastic loops are compared. Despite this, it is possible to say that an aquifer system exhibits mainly elastic behaviour (Zhang et al. 2014).

In contrast, when the effective stress in an aquifer system consistently exceeds the preconsolidation pressure due to a constant decline in piezometric levels, the aquifer undergoes significant plastic deformation, which is largely irreversible. However, in most aquifer systems, during the plastic deformation process, some seasonal partial recoveries occur, leading to unloading and partial recovery of the deformation. This behaviour is characterized by stress–strain curves that show a general subsidence trend (plastic deformation component) with overlapping elastic loops (recoverable deformation) (Burbey 2001; Zhang et al. 2007a, b, 2014).

Viscoelastic–elastoplastic behaviour involves a time-dependent component in addition to the characteristics of elastoplastic deformation. In this domain, the aquifer system exhibits a gradual deformation response over time, even under constant stress conditions (the groundwater level remains stable). The stress–strain curves in this domain demonstrate a progressive deformation trend that unfolds over extended periods and is influenced by the material composition and the stress history of the aquifer system (Burbey 2001; Zhang et al. 2007b).

Importantly, the specific behaviour of aquifer systems can vary widely on the basis of geological and hydrogeological conditions (Zhang et al. 2007a). Additionally, further research and field data are often needed to characterize and understand the behaviour of a particular aquifer system fully.

S-TOOL: design and implementation

To assess the effectiveness of the developed application, datasets from various aquifers around the world were chosen from the scientific literature, including those from Madrid (Béjar-Pizarro et al. 2017), the southern Yangtze Delta and the North China Plain (Zhang et al. 2014). Selected datasets were chosen according to data availability of both piezometric and ground motion, a representation of a variety of geological behaviours, and the existence of subjective analysis of the stress–strain curves by the researchers with which to compare our results.

Figure 1 summarizes the methodology applied in this paper by 2 S-TOOL. As shown in the figure, two parts are carried out by the algorithm. The first part, detailed in Sect. 2.2, focuses on automation for the construction and plotting of stress–strain curves. The second part, described in Sect. 2.3, was developed to automate the extraction and calculation of the S_{ke} and S_{kv} parameters from the stress–strain curves. Users may run both parts consecutively or, alternatively, run only the second part using the stress–strain curve as input.

Data collection and digitalization

To acquire the data utilized in this paper, it was essential to digitize time series plots for subsidence and piezometric measurements from aquifers in diverse global locations. The digitized datasets were subsequently employed to create stress–strain curves. Notably, the data used in the current study may differ slightly from those used in the original research from which they were obtained. These differences could arise from errors in the calibration of the time series axis, inaccuracies during the digitalization of points, or the selection of different points (i.e., a higher number or a lower number) than those used in the original time series during the digitalization process. It is important to note that these differences originate from the data reprocessing required for this study, rather than from the original research or the 2 S-TOOL code itself. The absence of access to the original numerical datasets from the source studies made it necessary to digitize time series from published figures to test the software. While this process can introduce minor inaccuracies, such as axis calibration shifts or point selection variability, these are inherent to the digitization procedure. In practical applications, where users work directly with their own original datasets, such potential sources of error would be substantially reduced.

Table 1 summarizes the information pertaining to the digitized time series data gathered from previous scientific studies conducted in various regions worldwide, which are described in Sect. 3.

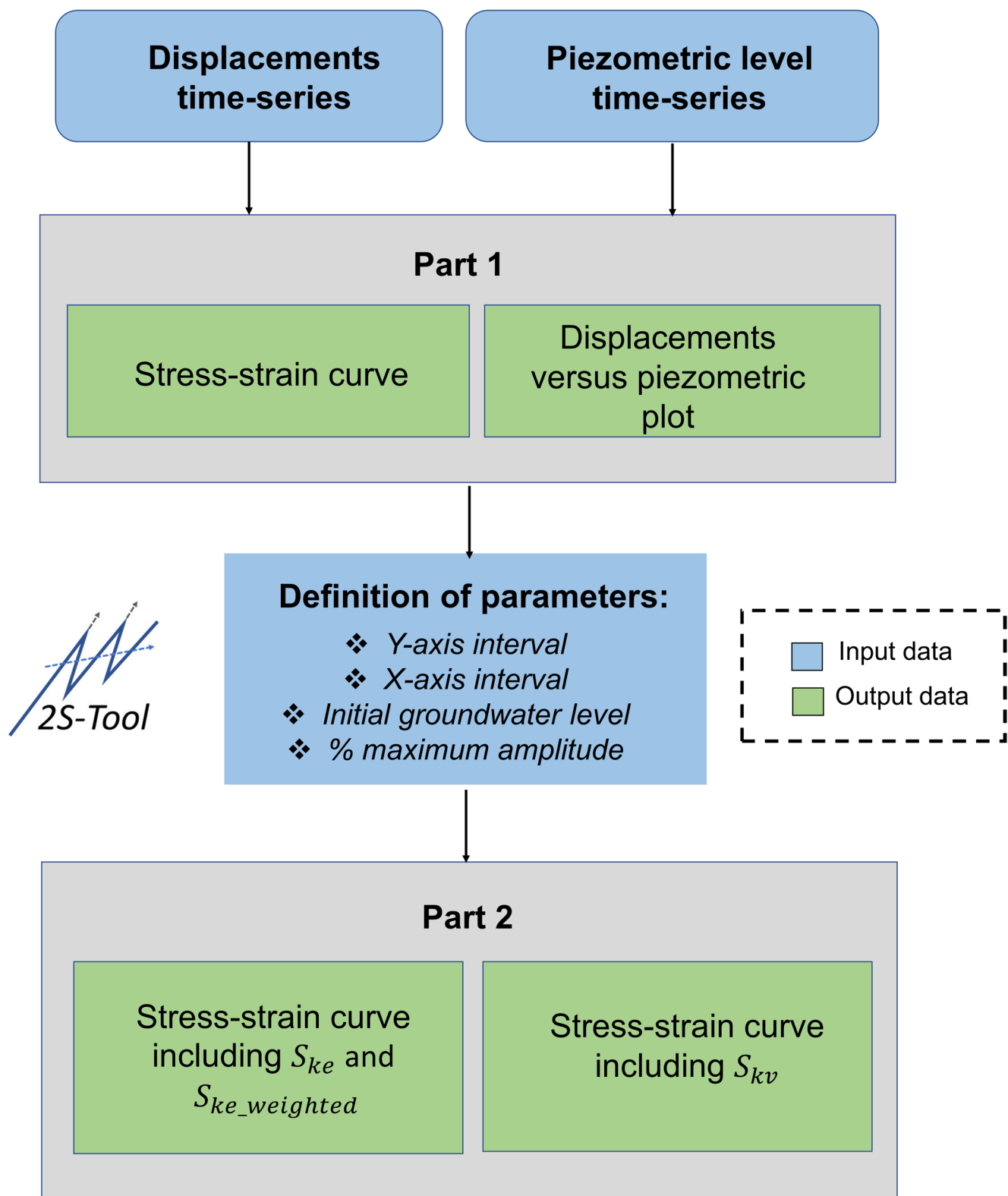


Fig. 1 Flowchart illustrating the process for the construction of stress–strain curves and the calculation of the S_{ke} and S_{kv} parameters performed by the developed application 2 S-TOOL

Table 1 Summary of digitized time series. E: Elastic; EP: Elastic; Ex: Extensometer

Reference	Location	N° of stress–strain curves	Instrumentation	Behaviour
Zhang et al. 2007a	Shanghai, China	3	Ex	EP
Zhang et al. 2007b	Shanghai, China	2	Ex	EP
Zhang et al. 2014	North China plain	7	InSAR	E and EP
Béjar-Pizarro et al. 2017	Madrid, Spain	17	InSAR	E
Li et al. 2022	North China plain	3	Ex	E and EP

Plotting the stress–strain curves

The first part of the developed open-access code 2S-TOOL focuses on plotting the stress–strain curves. The script of the algorithm can be found in the supplementary material and is accessible through the RUA repository (2S-TOOL, <http://hdl.handle.net/10045/150177>). In the first part, the program prompts the user to upload an Excel file containing a) prelisted time series of displacements (t_j , s_j), where ‘ s_j ’ represents the ground displacement measured at time ‘ t_j ’, with ‘j’ varying from 1 to ‘m’, and ‘m’ representing the number of displacement measurements; b) and groundwater depth time series (t_i , h_i), where ‘ h_i ’ represents the groundwater depth measured at time ‘ t_i ’, with ‘i’ varying from 1 to ‘n’, and ‘n’ representing the number of piezometric observations. These data are organized into four columns (a pair of columns accounting for the date ‘ t_j ’ and ground deformation ‘ s_j ’ values and an equivalent pair of columns for the date ‘ t_i ’ and the groundwater depth ‘ h_i ’) into an Excel template. Notably, the user is not required to input time series of the same length, initial and final dates or observation frequency. The program automatically identifies the common period from both time series, and if it exists, it continues with further analyses. Furthermore, the program automatically identifies the time format and uniformly determines the data format. The time series of deformation and groundwater depth are then represented in the same plot, using different vertical axes (piezometric depth in the left vertical axis and ground displacement in the secondary right vertical axis).

Notably, piezometric data must be input into the code as the groundwater depth measured from the ground surface. Since piezometric information is typically provided as elevation relative to sea level, it is necessary to convert groundwater elevation into groundwater depth via the following expression:

$$GD = GS - GW \quad (7)$$

where GD represents the groundwater depth, GS represents the ground surface elevation at the top of the piezometer and where GW represents the groundwater elevation.

This transformation is performed to avoid mathematical issues with the sign when calculating the slopes for determining the storage coefficient. However, if the ground surface elevation is unknown, users can set $GS = 0$, and therefore, $GD = GW$. In this last case, the piezometric data are expressed as the groundwater depth relative to an arbitrary reference plane. It is important to mention that for 2 S-TOOL surface elevation input is optional and used only to ensure consistent axis orientation in plots. This transformation does not affect the analysis outcomes. Additionally, a more detailed explanation of this parameter, including an illustrative figure, is provided in the user manual that accompanies the software package. This document is included in the code repository distributed with the present work. The final step involves generating the stress–strain curve. To develop this plot, common data from both piezometric (t_i , h_i) and ground displacement (t_j , s_j) time series are needed. Since the information usually comes from different sources and field campaigns, the data from both time series typically have distinct sampling dates (i.e., “ij” for the common period) and data acquisition frequencies. To address this issue, 2 S-TOOL first analyses and identifies the densest time series during the common period. Then, it extracts data from the densest time series using the dates from the other time series and performs linear interpolations between consecutive values. For example, considering that n_c and m_c are the number of data points in the common period for piezometry and ground displacement, respectively, if $n_c < m_c$, then the time series of displacements present a denser sampling compared with the groundwater depth time series; therefore, new values of displacement will be obtained by interpolation at the same dates when groundwater depth data exist. Once the interpolation process has finished, the application provides the data pairs of ground displacement and groundwater depth at the same dates, and then the stress–strain curves can be plotted via the interpolated pairs of s_i and h_i values calculated for the common dates. Linear interpolation between consecutive points of the time series was selected because it is robust and avoids overfitting artefacts in unevenly spaced datasets. Furthermore, since the time series data are usually spaced at most monthly, changes in trend between consecutive points are relatively small and can be reasonably approximated using linear interpolation. Nevertheless, users should be aware that interpolation can slightly influence slope estimates in the stress–strain curves, and the most reliable results are

obtained when the original datasets have high temporal coincidence and short sampling intervals.

Extraction of the S_{ke} and S_{kv} parameters from stress–strain curves

The second part of 2 S-TOOL focuses on extracting the skeletal storage coefficients of an aquifer system (elastic or inelastic) from the stress–strain curves built in part 1. The automation of this process involves the most complex step in the development of the application. The elastic skeletal storage (S_{ke}) can be calculated from the slope of the elastic segments identified in the stress–strain curve, as indicated in Eq. (6). Notably, this calculation is based solely on data from the increasing part of each slope and uses the loading branch of each cycle, which represents the deformation caused by groundwater decline. If the stress–strain curve reveals the existence of more than one elastic segment, the application will identify all of them and provide the S_{ke} for each cycle. It allows for the computation of a mean S_{ke} value while considering all S_{ke} values across cycles. This means that the S_{ke} value can be considered the characteristic value for the aquifer. One of the advantages of the proposed methodology is that the mean S_{ke} is based on all individual values of S_{ke} and not only the most significant values from the user's point of view. The value of the mean S_{ke} can be obtained by considering all elastic cycles identified by the application or applying a weighting criterion equally important. In the developed application, we consider a weighting for each elastic loop factor proportional to the magnitude of the loop. In doing so, the higher the elastic cycle is, the more significant it becomes in computing the mean S_{ke} of the aquifer. In any case, users can apply a different weighting criterion for each elastic cycle by themselves because the application includes the value of S_{ke} for each elastic cycle.

Similarly, inelastic skeletal storage (S_{kv}) can be calculated from the slope of the line that best fits all the groundwater depth–ground displacement data via Eq. (6). Finally, by knowing the thickness of the aquifer system or the analysed layer, it is possible to estimate the specific storage coefficient due to the compressibility of the solid skeleton (S_{sk}).

In detail, this part of the application starts by using the necessary data for generating the stress–strain curve described in Sect. 2.2, which is obtained by the first part of the 2 S-TOOL. The first column should contain ground displacement data as input, whereas the second column should contain groundwater depth data (Fig. 2a.). If users run part 2 of the code without running part 1, they must consider that both datasets of ground displacement and groundwater depth should be in the same length units (e.g., meters), and both correspond to the same -date. If both parts of the

2 S tool are run, the application will construct this information correctly automatically, and the application has been designed to identify any discrepancies in the amount of data between the two columns. In such cases, an error alert will appear on the screen, and the app execution will abort.

Defining the user-input parameters

To perform the analysis effectively and adapt the code to any case under analysis, the application requires four input parameters to be defined by the user (through a prompt window, Fig. 2b). These variables are as follows: (a) the 'Y-axis interval' (minimum variation in the piezometric level to accept the existence of an elastic cycle); (b) the 'X-axis interval'; (c) The preconsolidation depth, which indicates the minimum piezometric head (i.e., the maximum effective stress) recorded in the aquifer system throughout its history. It should be noted that this term is equivalent to the preconsolidation head but expressed as depth from the surface rather than as piezometric head.

Once the input data are correctly processed by the application, it automatically begins to identify between elastic and inelastic segments. To proceed with such a task, the user-defined parameters play a crucial role in identifying the segments and signals related to different mechanical behaviours. Their primary function is to enable the application's adaptation to any study case, considering the specific characteristics of each site, which may vary significantly (Table S1). Notably, properly applying and defining these parameters implies prior knowledge of the magnitude of ground displacement and piezometric heads in each study area. However, the application allows the user to test various parameter values and assess their suitability for a specific case study. The following section provides a more detailed description of each controlled parameter that must be set by the user prior to the calculation of the coefficients.

Y-axis interval

To distinguish elastic segments from inelastic segments, the relative maximum and minimum points on the stress–strain curve are identified. The maximum relative values (i.e., maximum groundwater depths) may indicate changes in aquifer system behaviour or even increase preconsolidation depth values. However, not all maximum points separate the elastic and inelastic segments. Therefore, it is essential to select those maximum values that genuinely mark the end of an elastic segment and the beginning of the next segment in the curve.

The application initially identifies all local maxima and minima, regardless of the magnitude of the cycle these max/min points are defining. However, many of the initially

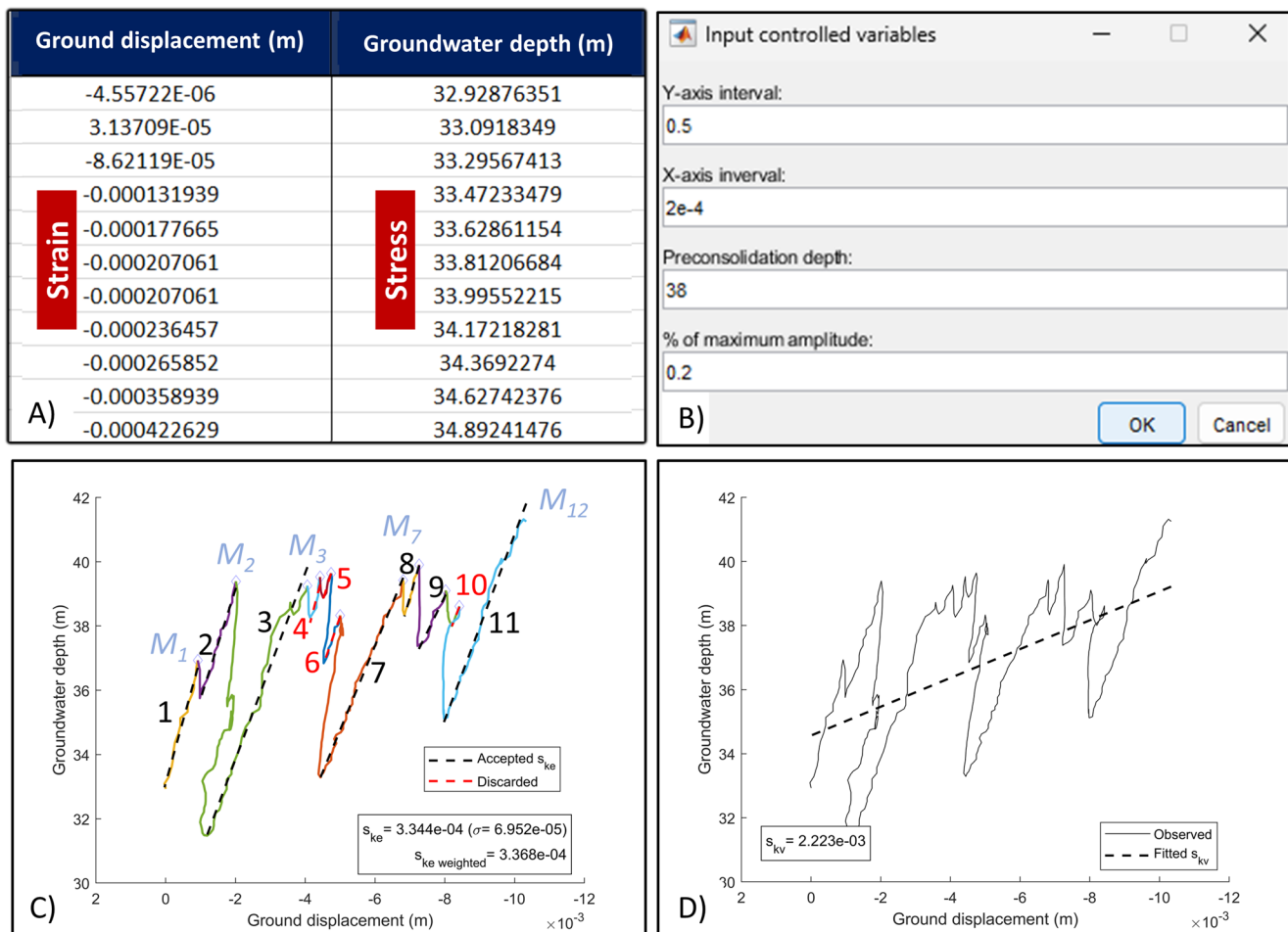


Fig. 2 (A) Excel template with the input data. (B) Screenshot of the dialog box for controlled variable input in MALAB. (C) Elastic cycles generated by the application (plotted in different colours). The valid elastic segments utilized in the analysis are denoted by 1, 2, 3, 7, 8, 9

and 11. Segments 4, 5, 6 and 10 were discarded because of their small amplitude. (D) Plot of the stress–strain curve including all the elastic loops. The dashed line represents the linear regression of the entire dataset

detected cycles may be very small and not significant compared with the whole context of the stress–strain curve. To address this situation, the first condition required to classify a segment as elastic is that the difference in the piezometric head between consecutive maximum and minimum points must exceed the value of the ‘Y-axis interval’ previously set by the user. When this condition is met, the relative maximum is saved. This process is repeated for all local maxima found in the curve, and the resulting values are stored in a matrix of maximum values.

After the relative maximum depths (M_i , with i varying from 1 to n) are identified and the preconsolidation depth (M_0) is defined, two different situations may arise when an elastic segment is defined:

- If the first maximum (M_1), which also satisfies the ‘Y-axis interval’ condition, has a lower piezometric value depth than the preconsolidation depth (M_0), M_1 will mark the end of an initial elastic segment. In this

scenario, the preconsolidation depth has not yet been exceeded, and the aquifer system (or the layer under analysis) exhibits elastic behaviour, as the piezometric head does not exceed the preconsolidation depth.

- Conversely, if the maximum (M_1) has a greater piezometric depth than that defined for the preconsolidation depth (M_0), there will be an elastic segment that will finish when the groundwater level reaches the preconsolidation depth, whereas the segment between M_0 and M_1 will be considered inelastic. Moreover, M_1 is then adopted as the new preconsolidation depth for further identification of elastic segments of the stress–strain curve.

This iterative process of comparison of successive maxima will be repeated for the subsequent maxima, enabling the identification of elastic and inelastic segments of the stress–strain curve.

The previously described logic operator is sequentially applied to all the maximum points that meet the ‘Y-axis interval’ condition. Importantly, the application prompts the user beforehand to specify the piezometric head changes they wish to consider for discretizing the elastic and inelastic segments.

X-axis interval

The second controlled condition involves setting a specific ‘X-axis interval’ value, which enables the mitigation of potential issues arising from the inherent distribution of data noise that can lead to artificial small and nonrepresentative segments. This condition specifies that to identify a point as a maximum, there must be a minimum horizontal gap between two consecutive maxima. Typically, the X-axis interval is quite small (e.g., 0.0001, 0.0002–0.005 m). The objective is to eliminate relative maxima that may arise from factors such as digitization, instrument accuracy, or even noise in the time series signal. Moreover, this condition allows the discarding of minor trends that might interfere with broader trends, which are the primary focus of this study (segments 4, 5, 6 and 10 shown in Fig. 2c). A good example where this criterion plays an important role appears in Fig. 3a, where the segment in blue is considered a

unique segment even though it shows very different maxima but with a small distance among them.

Preconsolidation depth

Another crucial aspect to consider during the interpretation of the stress–strain curves is the maximum historical stress preceding the analysed time series. This factor is represented in the proposed application by the ‘preconsolidation depth’. In most cases, this value is unknown in advance and is adopted as the first minimum value of the piezometric head of the time series. Consequently, if the initial data points of the stress–strain curve do not exceed this preconsolidation depth value, they are considered elastic segments. When the initial preconsolidation depth value is unknown, the application considers that the preconsolidation depth equals the first registered value of the piezometric depth of the time series. The first segment of the graph, if it shows an increase in the pressure value, is considered not to belong to any elastic domain until the curve reaches its first maximum, which satisfies the first two conditions mentioned above. This maximum will then replace the previous preconsolidation depth value and will be considered the starting point of the first elastic segment. A user may define a very high preconsolidation depth if he/she considers that the

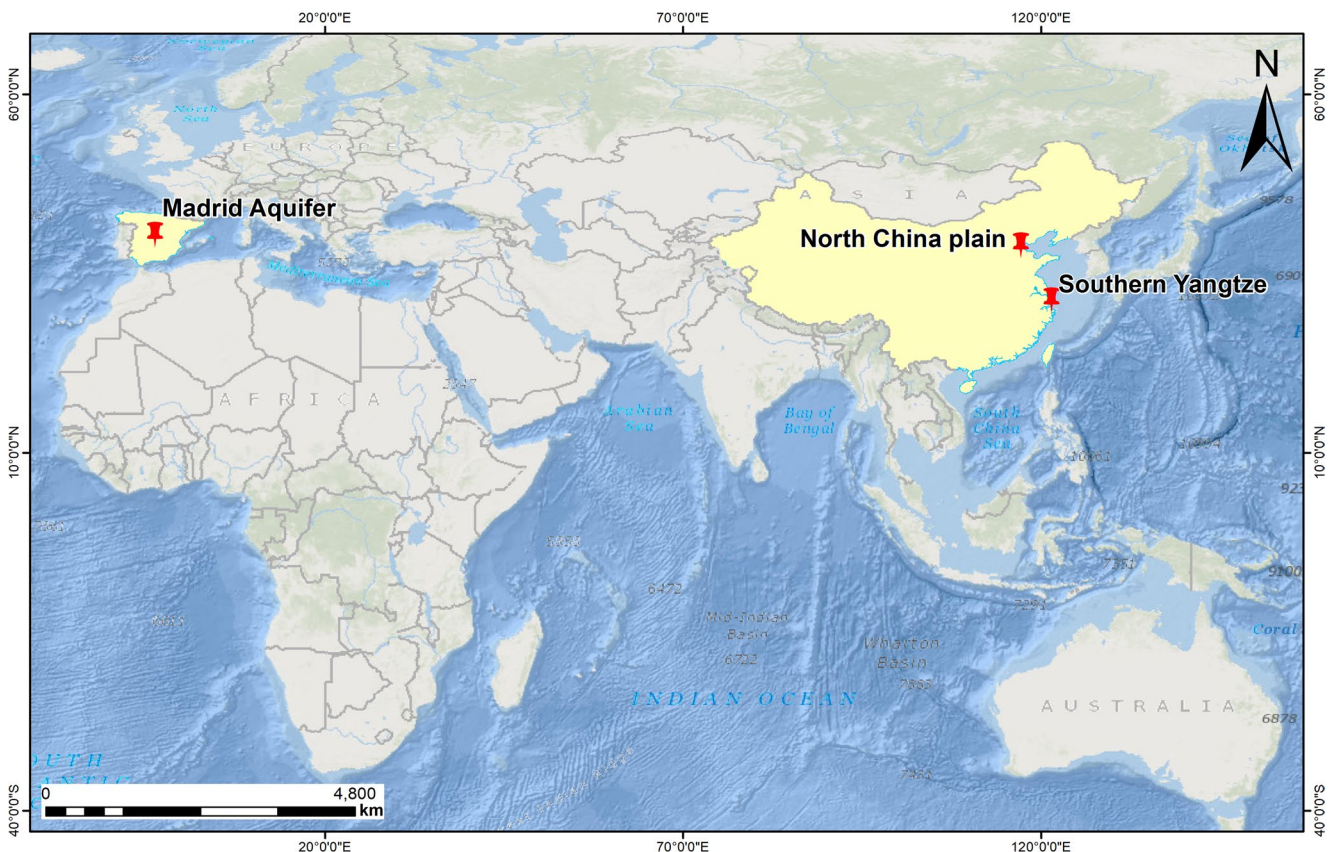


Fig. 3 Locations of the three case studies

stress–strain curve under analysis shows elastic behaviour from the beginning.

% of maximum amplitude

% of maximum amplitude

After delineating the various segments, the application performs a linear regression fitting for each elastic segment or loop identified as previously described. The inverse of the slope of each of these j segments (ground displacement/piezometric depth) provides skeletal storage (S_{kej}), specifically the elastic component. The mean value of S_{kej} across all the loops enables the determination of an average S_{ke} value for each stress–strain curve (Fig. 2c). Since the sizes of the loops can vary among them, it is also possible to estimate a weighted elastic skeletal storage ($S_{ke_weighted}$) by considering the total elastic points and the number of points composing each loop.

When examining the segmented stress–strain curves, it is common to observe elastic loops of considerable size alongside others that are relatively small. This situation can impact the results. Even though we consider the weighted value mentioned earlier, it is advisable to exclude very small loops from the calculation of S_{ke} . This is because linear fitting is much more likely to produce inaccurate estimations of S_{ke} from small loops than from the largest loops. To address this issue, the application identifies the loop with the highest amplitude in the Y-axis (i.e., with the difference between the maximum and minimum piezometric depths). Then, using the value of ‘% of maximum amplitude’, all small loops that have amplitudes smaller than the specified percentage of the maximum loop are ignored for further analysis. In practical terms, the code applies an amplitude percentage of the largest loop to all identified small loops to determine which loops to discard. For example, if the largest elastic loop has an amplitude of 10 m and the user-defined amplitude percentage is 20%, all loops with an amplitude less than 2 m will not be considered in the calculation of S_{ke} . Additionally, adjustment lines with an inverse slope are discarded, as they do not make physical sense (see Fig. 2c). This situation is mainly observed in very small loops, so the application of this last filter considerably reduces the number of loops exhibiting nonrepresentative or negative slopes.

The selection of the input parameters requires careful consideration and is not fully automatic. High values of Y-axis interval or X-axis interval reduce the number of elastic loops detected by the algorithm, as they impose stricter constraints on their identification. The parameter preconsolidation depth can usually be defined from historical groundwater level records or by selecting the

maximum observed value in the series when the historical information is not available. The percent parameter controls the exclusion of very small loops, which may not be representative of the aquifer system response. Users who wish to include these loops can set a lower value for this parameter.

In practice, some additional remarks can help in the adjustment of these parameters. If some segments of the curve are missing, this usually means that the value of the preconsolidation depth should be increased. When two loops appear or are detected as a single one, it is convenient to decrease X-axis interval in order to separate them. If the regression line obtained for a loop shows a slope that is lower or more horizontal, a higher value of Y-axis interval is recommended to obtain a more consistent fit.

As a general guideline, the loop parameter can be set in the order of the seasonal component of the time series, which often provides a reasonable starting point for analysis. Thus, the procedure remains supervised and requires expert judgement, which ensures consistency of results while acknowledging the subjective component involved in parameter selection.

Graphical results

The algorithm produces an initial graph showing the groundwater depth versus ground displacement as the output. This graph encompasses all the differential elastic segments defined by considering the first three user-controlled variables: the ‘Y-axis interval’, the ‘X-axis interval’, and the preconsolidation depth prior to the analysed time series. Each loop is represented by its best-fit line for the data, indicated by a dotted line. The black adjustment lines represent the slopes used to calculate the elastic skeletal storage. Conversely, the red lines indicate segments that have been discarded either because they do not meet the ‘% of maximum amplitude’ criterion and/or because the slope of the line is inverted (see Fig. 2c). Afterward, the algorithm generates a second graph displaying the complete stress–strain curve data in black. The elastic segments defined earlier are overlaid with these data in a different colour. As a result, the inelastic segment is represented by a black dashed line. Finally, the linear regression of the complete dataset allows for the calculation of the inelastic skeletal storage coefficient (S_{kv}) by determining the inverse slope of the adjustment line (see Fig. 2d).

In addition to generating the graphics mentioned above, the application also saves on the last sheet of the input Excel templated all S_{ke} , $S_{ke_weighted}$ corresponding to each loop (both selected and discarded) as well as the S_{kv} .

Overview of case studies used for 2 S-TOOL validation

This section provides a concise description of the geographical, geological, and hydrogeological characteristics of the three case studies (2 in China and 1 in Spain) used (see Fig. 4) to evaluate the novel application. The main aim is to provide the hydrogeological environmental background of the dataset, and the results used to validate the outcomes obtained from 2 S-TOOL.

Southern Yangtze Delta, China

The southern Yangtze Delta, characterized by its extensive flat terrain, is located within three economically significant provinces in China. The notable economic and industrial advancements in this region have generated an increased demand for groundwater, subsequently leading to substantial land subsidence (Zhang et al. 2007a). The region hosts a highly complex aquifer system comprising Quaternary materials derived from fluvial, lacustrine, and oceanic environments. These materials include gravel, sand, silty sand, silty clay, and clay. The depth of the aquifer system varies from 250 to 310 m beneath Shanghai, but in certain areas, it can exceed 360 m (Zhang et al. 2007b). The system comprises five aquifer layers that are primarily composed of granular materials such as gravel and sand. The topmost aquifer is unconfined, whereas the remaining four aquifers are confined. These aquifers are separated by lower-permeability materials such as silt and clay, corresponding to aquitard layers.

In this territory, land subsidence was first reported in 1921 in Shanghai. The first instance of land subsidence due to groundwater extraction was approximately 24 mm/year. However, after 1949, when groundwater pumping significantly increased, land subsidence reached values of up to 110 mm per year. At that point, authorities implemented measures to mitigate land subsidence, including increasing artificial recharge and relocating groundwater extraction to deeper confined aquifer layers. Despite these efforts, the declining trend in piezometric levels did not cease until the end of 1999. As a result, three cones of depression from three different important cities were joined, forming a unique and large regional cone of depression covering an area of 7,000 km², with groundwater levels decreasing by 87.7 m in the centre of the cone. By 2000, the maximum cumulative subsidence in the area had reached 2.8 m (Xue et al. 2008).

North China plain (Beijing City & pinggezhuang and Tianzhu stations)

Beijing is renowned as one of the most densely populated cities in the world and is classified as having one of the highest subsidence rates in China (Syvitski et al. 2009). The

vast plain where Beijing is located covers an area of almost 6,400 km² and is delimited by the Taihang and Jundu Mountains. The sediments constituting this plain originated from fluvial and alluvial processes. These deposits have led to the development of a distinct aquifer system with well-defined units. The bedrock consists of Cambrian–Permian rocks and forms the lateral and lower boundaries of the aquifer. Overlying the bedrock are volcanic rocks from the Palaeogene–Neogene period. The sediments that fill this rock geometry display varying granulometry and date back to the Pliocene and Quaternary periods, corresponding to the fluvial deposits of the delta (Zhang et al. 2014). The aquifer system is a multilayer system consisting of five confined aquifers separated by low-permeability layers. Currently, the three uppermost aquifers are the most heavily exploited, serving as sources of water for domestic and industrial purposes.

The Tianzhu extensometer in northern Beijing, located at the confluence of the Wenyu and Chaobai River alluvial fans, has monitored an area of intense groundwater extraction since 2004, with over 500 m of multilayered aquifer deposits. The Pinggezhuang extensometer, positioned in one of the highest subsidence zones (up to 24 mm/year) on the Chaobai River alluvial fan, records 380 m of heterogeneous Quaternary deposits. Both stations lie within sedimentary formations from the Palaeogene to Quaternary periods, shaped by fluvial and alluvial processes (Li et al. 2022).

The long-term overexploitation of groundwater has caused a drastic decline in piezometric levels, with a maximum decrease of approximately 30 m in both phreatic and confined aquifer systems from 2004 to 2010 (Zhu et al. 2015). This significant reduction in groundwater levels resulted in sediment compaction and extensive land subsidence. The most important cumulative land subsidence in the region reached 600 mm from 1955 to 2009. To monitor these displacement boreholes, extensometers were set up in 2004 and 2008 at several sites, with the deepest monitoring reaching approximately 250 m and a reference point established at a bedrock depth of 900 m. Furthermore, from June 2003–January 2010, the permanent scatterer interferometry (PSI) study measured the greatest annual land subsidence, reaching 52 mm/year, with a maximum cumulative subsidence of 342 mm (Zhu et al. 2020).

The South-to-North Water Diversion Project's central route commenced operations in December 2014, aimed at supplying water to Beijing to mitigate groundwater extraction and curb land subsidence. Recent studies have highlighted the effectiveness of the project, revealing a significant reduction in land subsidence in the region (Du et al. 2021; Lei et al. 2023).

Madrid aquifer, central Spain

The Madrid Tertiary aquifer system is a sedimentary basin covering an area of approximately 600 km². This basin

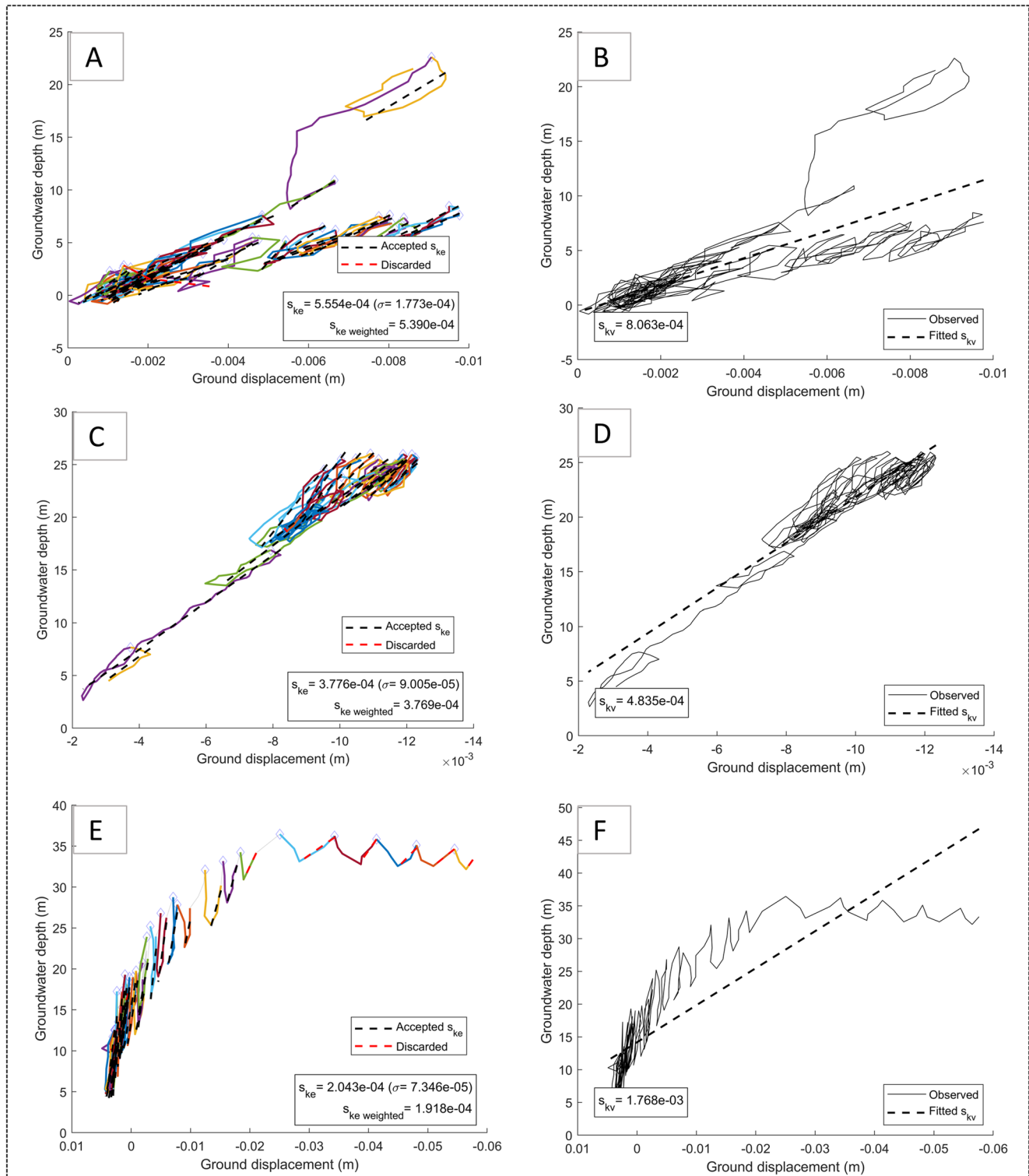


Fig. 4 Analysis of stress–strain curves from the Southern Yangtze Delta. (A–B) Second confined aquifer; (C–D) Second aquitard layer. (E–F) Third aquitard layer

presents a triangular shape and is surrounded by various mountain ranges, including the Toledo, Altamira, and Central System Ranges. The aquifer in this region corresponds to an unconfined aquifer system comprising materials from continental environments dating from the early Cenozoic to the upper Miocene. This reservoir represents the primary freshwater reservoir in the Community of Madrid. While the mean thickness of the aquifer typically measures approximately 1,000 m, in certain locations, it can extend to a potential depth of 3,000 m. The aquifer's multilayer structure consists of different levels of sand lenses embedded within a low-permeability clay matrix (Béjar-Pizarro et al. 2017).

The analysis conducted by Ezquerro (2021) identified two distinct subsidence zones in proximity to well fields. During the ERS period (1992–2000), subsidence occurred around the wells during the extraction periods, with uplift during recovery. Three subtly distinct behavioural patterns were identified: (a) In Fuencarral, subsidence followed a cyclical pattern with up to 5 cm of displacement, correlating with extraction periods; (b) Pozuelo de Alarcón experienced more continuous subsidence with less recovery; and (c) the results also revealed that piezometric drops of 120–200 m caused up to 8 cm of subsidence and uplift in different cycles, with significant impacts within 3 km of the wells, decreasing to negligible levels at 7 km. Subsidence in Fuencarral nearly fully recovered because of an elastic response linked to sandy deposits, whereas Pozuelo de Alarcón's slower recovery was attributed to the clay content, which caused elastoplastic behaviour. The extraction and recovery timing directly influence the extent of subsidence recovery in each cycle (Ezquerro 2021).

Results

This section details the results for each case study described in Sect. 3. The parameters used for the calculation of the coefficients are listed in the supplementary material (Table S1).

Southern Yangtze Delta, China

Five stress–strain curves from the aquifer beneath Shanghai city were analysed, each corresponding to different extensometers and/or different aquifer layers. From the information provided in Zhang et al. (2007a), three stress–strain curves were constructed. The first curve corresponds to the second confined aquifer. In this curve, the application identified 39 elastic segments; however, for the calculation of S_{ke} , only 37 of these segments were used since two loops were automatically discarded because they had a negative slope, as shown in Fig. 5a. The second stress–strain curve corresponds to the second aquitard, and it exhibits loading and unloading cycles due to seasonality (Fig. 5c). The elastic

skeletal storage was calculated via 37 elastic loops, and the adjustment curve that encompasses all the data aligns well with the general trend of the stress–strain curve (Fig. 5d).

The last curve corresponds to the third aquitard (Figs. 5e–f). In this curve, it is evident how the loops widen as subsidence increases, leading to a progressive increase in the slope of the line. However, the last five elastic segments exhibit opposite slopes, which is why these segments were automatically discarded. As shown in Fig. 5f, calculating a single elastic and inelastic coefficient for the entire curve would not accurately represent reality, as this layer exhibits different behaviours depending on the considered period. The behaviour in the first part of the curve is predominantly elastic, whereas in a later period, it exhibits elastoplastic behaviour. Finally, towards the end of the curve, the widening loops indicate that compaction continues even when the water level stabilizes, suggesting a time lag in deformation. This behaviour implies viscous properties within the aquitard, where compaction persists for two possible reasons: the slow dissipation of pore water pressure caused by the lower permeability of the aquitard and creep deformation under constant stress. These characteristics reflect a transition from elastic or elastoplastic behaviour to viscoelastoplastic behaviour as groundwater levels decrease below previous minimum values (Zhang et al. 2007a). Therefore, we calculate the coefficients in segments, dividing the stress–strain curve into different parts, to compare the discrepancies with those of the author, who also performs the calculation in segments. The results obtained were as follows: the value of S_{ske} obtained via 2 S-TOOL for the first part of the curves was $6.76 \times 10^{-6}/m$, whereas the author value was $6.61 \times 10^{-6}/m$. For the elastoplastic period domain, the value of S_{skv} obtained via 2 S-TOOL was $2.43 \times 10^{-5}/m$ greater than the value calculated by Zhang et al. (2007a), which corresponds to $4.38 \times 10^{-5}/m$. This example highlights that, in cases where the behaviour of the aquifer changes over time, it is necessary to analyse the stress–strain curves by segments previously identified by the operator.

The curves analysed in Zhang et al. (2007b) were also evaluated via 2STOOL. These curves correspond to additional levels of the same aquifer, and the results obtained are presented in the supplementary material (Figure S1). Since the thicknesses of the layers (aquitards and aquifers) are known, it was possible to calculate the elastic and inelastic specific storage coefficients (S_{ske} , $S_{ske_weighted}$, and S_{skv}).

North China plain (Beijing city)

Seven stress–strain curves were obtained from Zhang et al. (2014); these curves correspond to three aquifer levels and four aquitard levels. In the present section, we select and describe only one aquifer and one aquitard layer. The stress–strain curves for the remaining layers, which exhibit

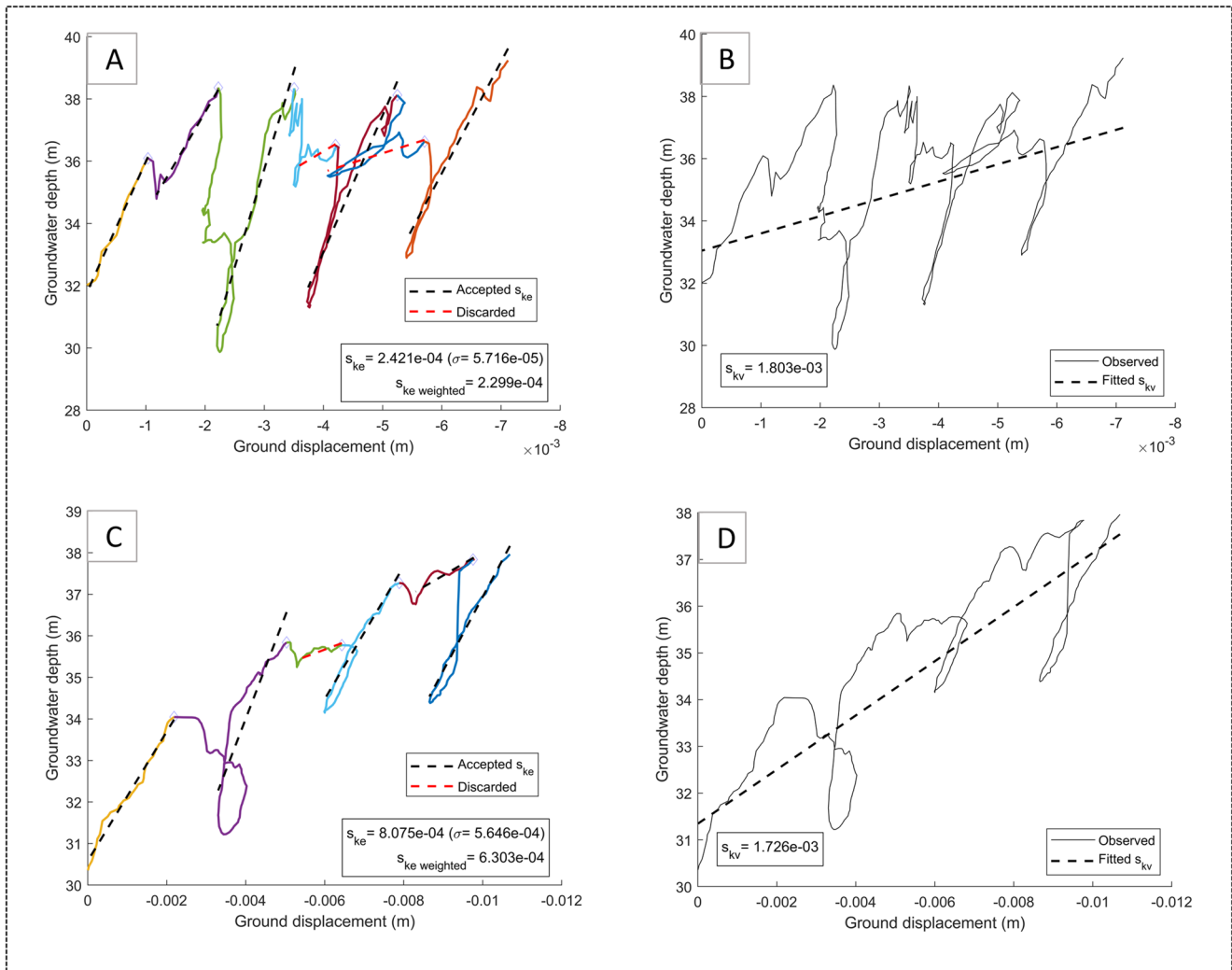


Fig. 5 Analysis of stress–strain curves from the Plain of Beijing. (A–B) Second aquifer layer; (C–D) last aquitard layer

trends similar to those shown in this section, are presented in supplementary material Figures S2–S3.

At the second aquifer level, seven elastic segments were identified by the application (Fig. 3a). In this case, out of the five segments identified by the code, two were discarded because of their negative slopes. This condition was sufficient to exclude the segments shown in blue and light blue in Fig. 3a. These segments had adjustment lines that did not correspond to logical values for the calculation. As a result, the standard deviation of the data exhibited small magnitudes, on the order 10^{-5} and 10^{-4} . Importantly, three segments significantly influence the coefficient calculation. The first segment in green comprises 44 data points, whereas the segment in red has 55 points, and the last segment in orange has 49 points. The first segment of the stress–strain curve in blue was considered an elastic segment. This classification was based on the third variable controlled by the user, which is the initial preconsolidation depth. Finally, Fig. 3b

displays the fitting line for all the data, illustrating that the global trend aligns well with the entire set of elastic loops.

The last aquitard layer corresponds to the maximum depth of the system, and this layer clearly tends to increase in ground displacement (subsidence) as the groundwater depth increases. Figure 3c illustrates the presence of four elastic loops that display similar values of the adjustment lines, resulting in low standard deviation values. Notably, the adjustment line for all the data points closely matches the general trend of the stress–strain curve (Fig. 3d).

North China plain (Pinggezhuang and Tianzhu stations)

This section presents three of the six stress–strain curves analysed by Li et al. (2022), which represent the elastic, elastic–inelastic, and inelastic behaviours of different aquifer system layers.

The stress–strain curve corresponding to the layers analysed by the Tianzhu extensometer is depicted in Fig. 6a. In this stress–strain curve, six elastic segments were initially identified. However, one of them was excluded because it did not meet the percentage of the amplitude condition, which was set at 20%. Additionally, this small loop clearly has an incoherent S_{ke} value compared with the other five loops. Consequently, five segments were considered for calculating the storage coefficients. These five segments exhibited similar values, resulting in a low standard deviation. Like the previous stress–strain curves, the fitted line closely followed the overall trend, passing through all major elastic loops (refer to Fig. 6b).

One of the layers analysed from the Pinggezhuang station dataset predominantly exhibits elastic behaviour, as illustrated in Fig. 6c. In this layer, eight elastic loops were identified; however, one loop was discarded because of

its inverted slope. Among the loops, one is recognized as the most representative, comprising 45 data points. Overall, the loops exhibit similar slope values, all within the same order of magnitude. Owing to the predominantly elastic behaviour of this layer, it was not necessary to calculate S_{kv} .

Another layer studied by the Pinggezhuang extensometer shows a stress–strain curve that is notably different from those of the other layers in the study area. As shown in Fig. 6d, this layer does not display a recovery of cumulative subsidence when the piezometric level decreases. Since there is not a hysteric loop, this indicates that the deformation is mainly inelastic; for comparison purposes, the elastic value is not considered in the validation, following the values provided by the author. In this case, the S_{kv} value shown in Fig. 6d is used solely for the validation analysis in the discussion section.

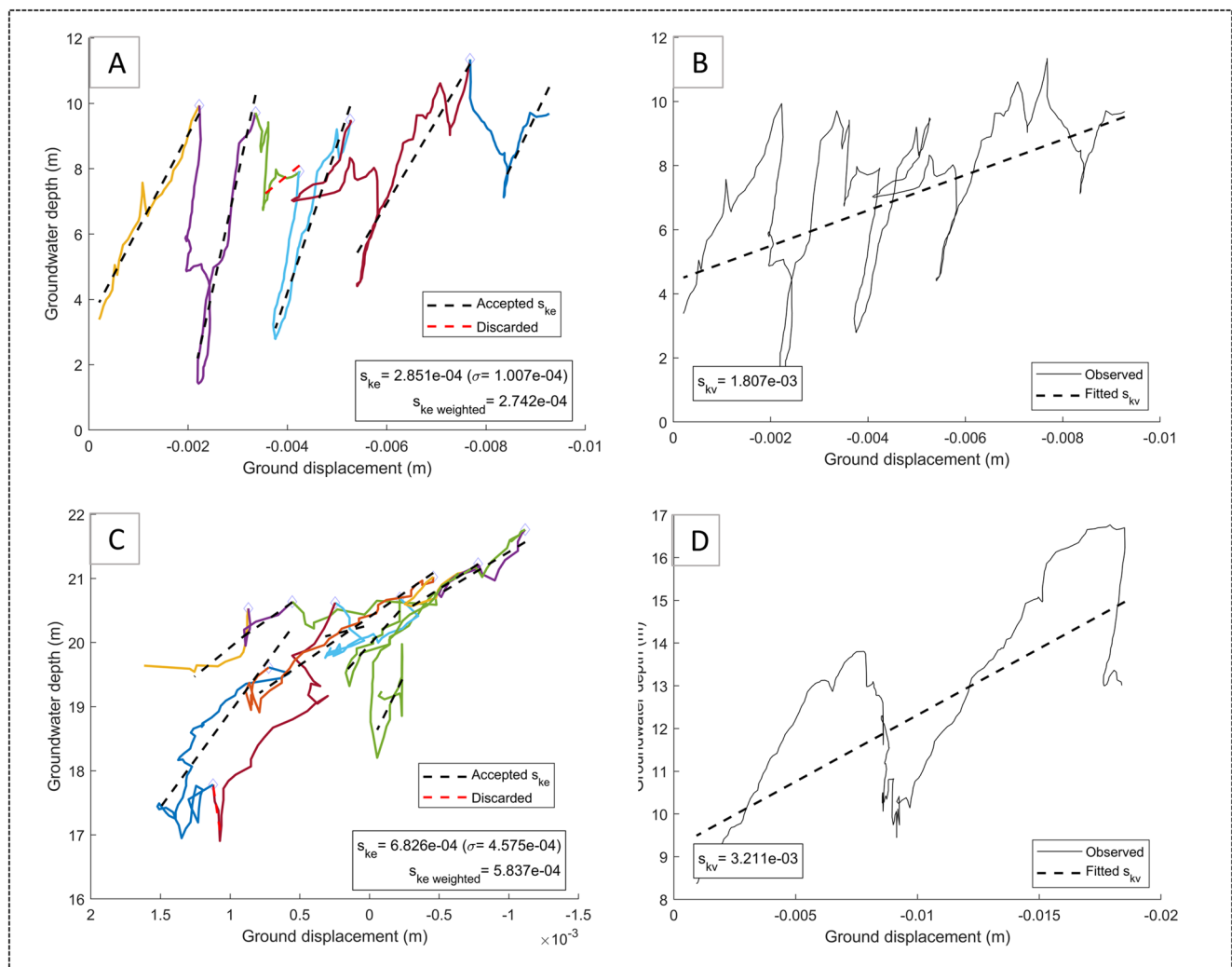


Fig. 6 Analysis of the stress–strain curves from the North China Plain. (A–B) Second layer - Tianzhu station; (C–D) First and third layers - Pinggezhuang station

Madrid aquifer, Central Spain

In this study area, 17 stress–strain curves digitized from the supplementary material of the paper published by Béjar-Pizarro et al. (2017) were built on the basis of the different wells distributed in the Madrid Tertiary aquifer system. To obtain the storage coefficients of the aquifer system.

These curves exhibit notable differences compared with those previously analysed. In this case, the time series are relatively shorter, spanning from 2004 to 2010. Additionally, the intrinsic characteristics of the materials result in a single, distinctive elastic loop in most cases without the observation of inelastic deformation. Figure 7 shows some of the stress–strain curves generated by the application of 2 S-TOOL. All the other curves are shown in Figures S4–S5.

A similar overall trend is observed in most of the stress–strain curves, with the presence of an important elastic loop. Across all the curves, two primary segments can be identified. One of these segments is associated with the loading and unloading branches of the main loop. Furthermore,

in cases where curves exhibit smaller loops, they typically consist of more than two larger elastic segments. These smaller loops tend to appear at the edges of the largest loop (see Fig. 7).

Because the detrital material that makes up the Madrid aquifer primarily exhibits elastic behaviour, there is no need to calculate the inelastic skeletal storage coefficient (S_{kv}). Instead, the S_{ke} and $S_{ke_weighted}$ coefficients are included.

Discussion

In this section, the storage coefficients obtained via the 2 S-TOOL application are validated by comparing them with those calculated by the authors of the respective studies via expert criteria (i.e., manual fitting) on the basis of the methodology proposed by Riley (1969). Notably, the main objective of the proposed application is not to provide a tool able to reproduce the results obtained manually but to offer a new methodology whose results may differ slightly from

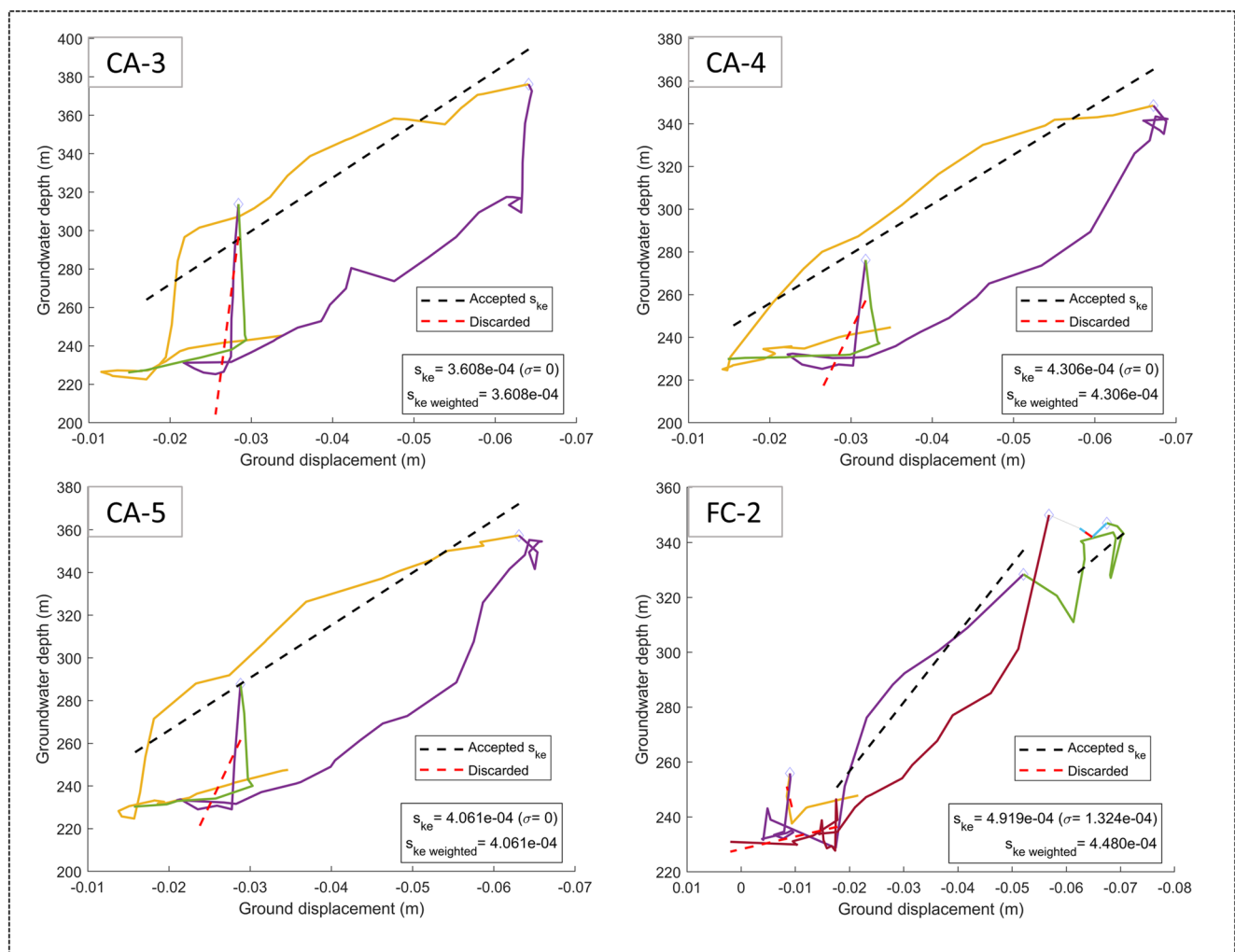


Fig. 7 Analysis of the stress–strain curves from the Madrid Tertiary aquifer system

those obtained following subjective observations. Nevertheless, it is obvious that both results may not differ significantly since they analyse the same input data.

In the southern Yangtze Delta Basin, all the stress–strain curves exhibit elastoplastic behaviour, and some of them present a viscous component, involving both recoverable and nonrecoverable deformations. The storage coefficients calculated via 2 S-TOOL are very similar to those calculated by Zhang et al. (2007a, b). Furthermore, for all the coefficients, the S_{ke} value calculated by Zhang et al. (2007a & 2007b) falls within the range of the maximum and minimum slopes identified by the application and is relatively close to the average S_{ke} . This indicates that $S_{ke_Zhang(2007)}$ corresponds to one of the elastic loops considered by the application (see Fig. 8). Importantly, the discrepancies are related primarily to the fact that Zhang et al. (2007a & 2007b) calculated the coefficients on the basis of only one elastic loop, whereas the application is capable of independently considering all the elastic loops and subsequently calculating statistical parameters such as the mean and weighted mean of S_{ke} . This approach aims to increase accuracy and reduce subjectivity.

All the S_{kv} values closely resemble those calculated by Zhang et al. (2007a, b). These values are presented in Table S2, which displays all the coefficients calculated by the application alongside those computed by Zhang et al. (2007a, b) for each aquifer and aquitard layer. A comparison of the S_{ke} values is also depicted in Table S2.

In the plain of Beijing, only the first confined aquifer layer has elastic behaviour, and the rest of the layers present

elastoplastic behaviour. All the elastic and inelastic skeletal storage coefficients obtained by the application of the developed application are shown in Table S3.

In the case of the shallow aquifer layer of the plain of Beijing, which exhibited elastic behaviour, similar S_{ke} values were obtained with the developed application compared with $S_{ke_Zhang(2014)} 2.296 \times 10^{-4}$ and 2.72×10^{-4} . In this study, $S_{ke_Zhang(2014)}$ did not consistently fall within the range defined by the maximum and minimum slopes of the other coefficients, indicating discrepancies (Fig. 9). Moreover, $S_{ke_Zhang(2014)}$ tended to have numerically smaller values than did S_{ke} and $S_{ke_weighted}$, which was due to the asymmetrical behaviour of elastic loops in the curves (Fig. 9), since in the proposed methodology, only the segment where deformation increases with groundwater depth is considered for calculating the elastic storage coefficient, whereas Zhang et al. (2014) fit an average line encompassing both the loading and unloading asymmetrical segments. Another important factor contributing to the differences in the results was the number of segments considered to estimate the elastic storage coefficient. While Zhang et al. (2014) calculated this parameter via a single line, our methodology considers (S_{ke} , $S_{ke_weighted}$) of the coefficients.

In summary, the S_{kv} coefficients in the curves generally exhibited a similar order of magnitude to those in Zhang et al. (2014) (Table S3). Any minor variations could be attributed to the digitization process of the results from Zhang et al. (2014), which were used in the present study as input

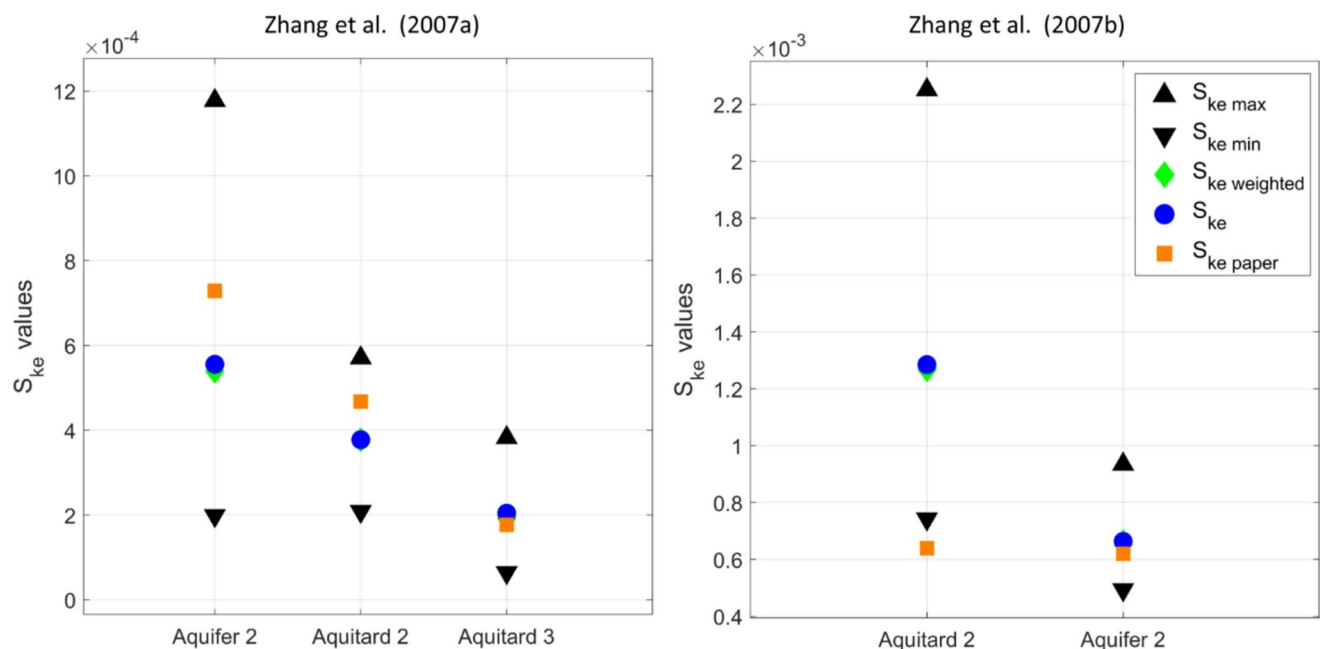


Fig. 8 Comparison between the elastic skeletal storage coefficients calculated by 2 S-TOOL and the coefficients calculated by Zhang et al. (2007a, b)

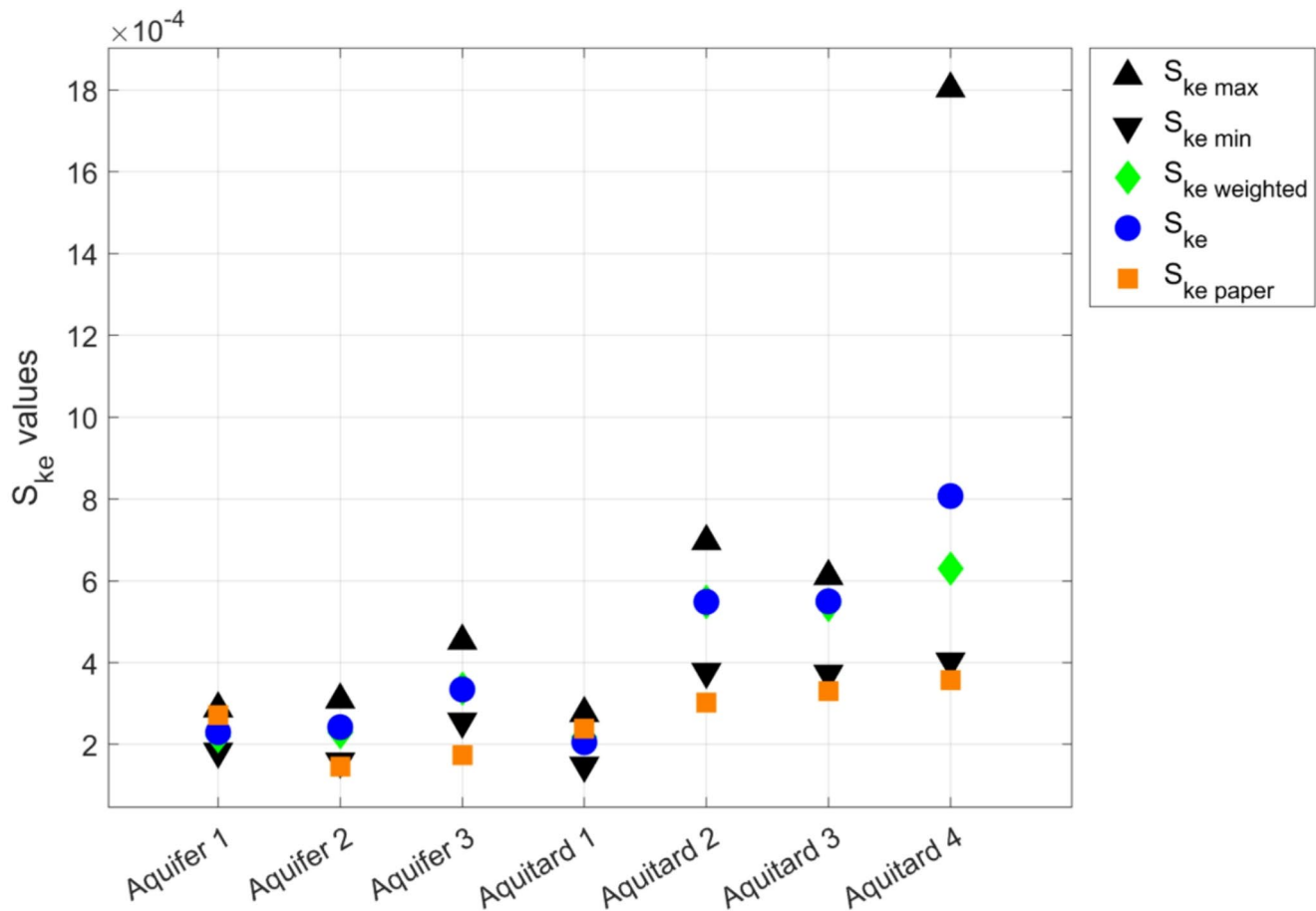


Fig. 9 Comparison between the elastic skeletal storage coefficients calculated by 2 S-TOOL and the coefficients calculated by Zhang et al. (2014) in the plain of Beijing

data for the developed application. Notably, the aquitard levels displayed a slightly distinct global fitting line compared with Zhang et al. (2014), being somewhat more vertical and producing higher S_{ke} values than Zhang et al. (2014).

With respect to the Pinggezhuang and Tianzhu stations in the North China Plain, the study reveals three distinct curve behaviours: elastic, elastoplastic, and primarily plastic. In Table S4. The storage coefficients S_{ke} and $S_{ke_weighted}$ and S_{kv} closely align with those reported in Li et al. (2022).

In the Pinggezhuang extensometer, the first curve demonstrates elastic behaviour, and the four identified elastic segments exhibit similar trends, producing S_{ke} and $S_{ke_weighted}$ values near $S_{ke_Li(2022)}$ (6.864×10^{-4}). Notably, $S_{ke_Li(2022)}$ consistently falls within the range defined by the S_{ke_max} and S_{ke_min} mean values of these coefficients (Fig. 10). In the second layer of the Pinggezhuang extensometer, $S_{ke_Li(2022)}$ does not align with the maximum and minimum values from the analysis. Nevertheless, the numerical values are strikingly similar, with all the S_{ke} values falling within the same order of magnitude (Table S4; Fig. 10).

Finally, despite the absence of any elastic behaviour, the layer studied by the Tianzhu extensometer was also analysed via 2 S-TOOL. This emphasizes the significance of continuous oversight by a qualified researcher throughout the analysis, especially when controlling for variables and interpreting results. In this scenario, as deformation does not rebound as the water level increases, elastic segments are not distinguished. Instead, it identifies periods where deformation halts or slows without recovery.

For the Madrid aquifer, that study combined InSAR data and water level measurements to obtain stress–strain curves that exhibit elastic behavior. Unlike other study areas that focus on specific aquifer layers, this research aims to assess the overall behaviour of the multilayer aquifer system. Among the 16 curve analyses, the average of all elastic storage coefficients and the mean values taken from Béjar-Pizarro et al. (2017) are presented in Fig. 11. The application-derived values with the novel application closely resemble the Béjar–Pizarro et al. (2017) average.

The main difference observed in the case of the Madrid aquifer compared with the previously analysed cases is that in

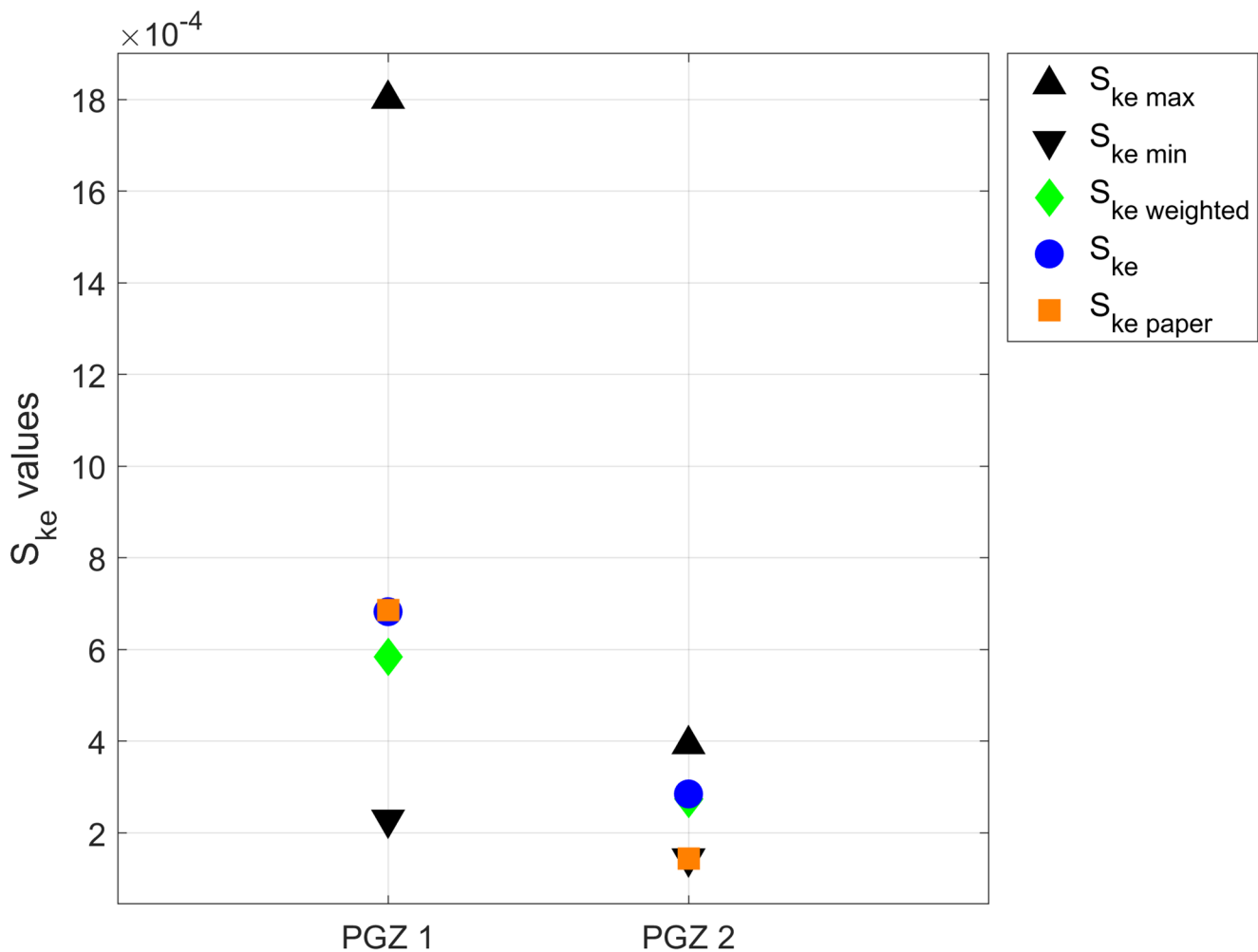


Fig. 10 Comparison between the elastic skeletal storage coefficients calculated via 2 S-TOOL and the coefficients calculated by Li et al. (2022) in the North China Plain

most of the curves, only the main loop was considered, ignoring minor or secondary loops. This is because primary cycles exhibit a significantly larger loop than secondary cycles do and are assumed to be the most representative for the calculation of S_{ke} . This is the main reason why, in Fig. 11, most wells do not display a value for S_{ke_min} , S_{ke_max} , or $S_{ke_weighted}$. However, some curves do exhibit secondary loops that are considered significant apart from the main loop (see FC-2 in Fig. 7). In these cases, more than one S_{ke} value was considered for the calculation, as shown for wells CB-9, FA-3, FC-2 and MJ-1. The curves corresponding to these examples are included in Figures S4 and S5. The user may control the rejection of minor loops by using the “% of maximum amplitude” parameter.

In closing, it is noteworthy that this work does not aim to replicate the exact results but rather demonstrates the ability to analyse stress–strain curves with elastic and plastic behavior. Notably, the obtained elastic storage coefficients, alongside $S_{ke_Béjar(2017)}$, reveal consistent results, falling

within an order of magnitude of 10^{-4} in all the cases (Table S5).

Finally, a comparison was also made between the S_{kv} values calculated by 2 S-TOOL and those reported by various authors, which are summarized in Fig. 12. This figure shows that, in general, the values are quite similar. In some cases, the results match almost exactly with those provided by the authors, as seen for aquitard 3 in Zhang et al. (2014). As previously mentioned, S_{kv} graphically corresponds to the slope of the line that best fits the stress–strain curve. However, in certain specific cases, significant discrepancies were found, such as for aquitard 2 in Zhang et al. (2014). In this case, the authors calculate the slope of a line that originates from the first identified loop to the last, but this loop does not coincide with the first segment of the curve. Conversely, 2 S-TOOL considers the first elastic segment starting from the beginning of the curve (see Figure S3b), which results in a steeper slope calculated by the author, whereas the slope calculated by 2 S-TOOL is more gradual.

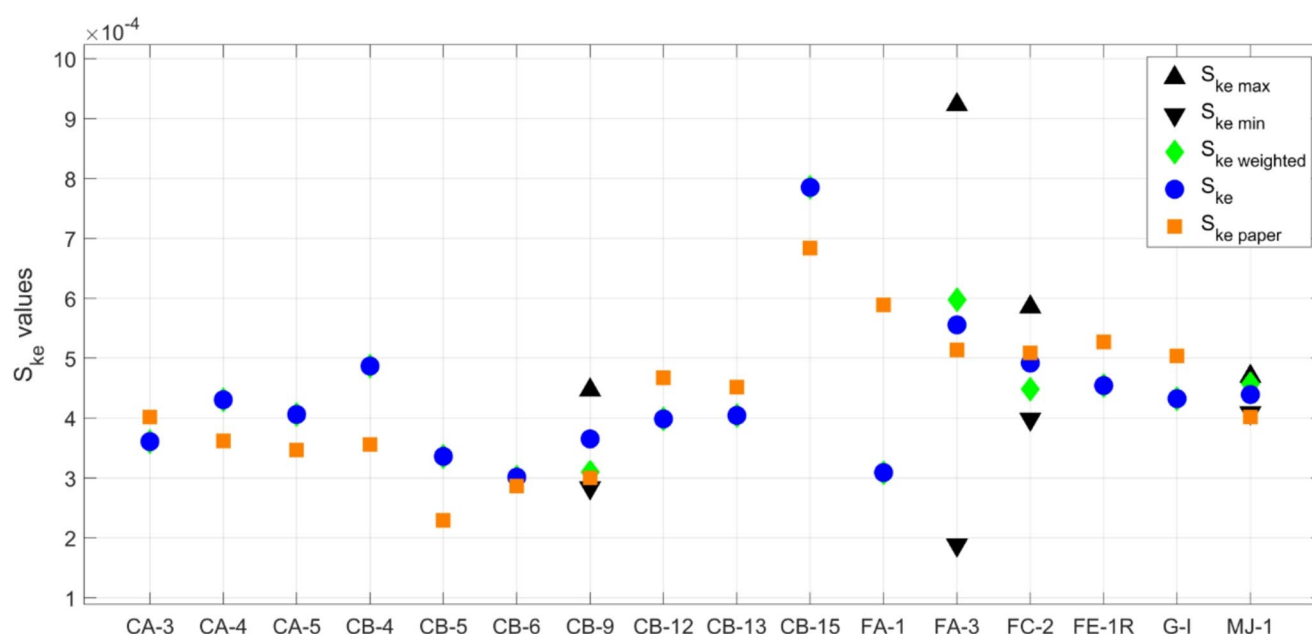


Fig. 11 Comparison between the elastic skeletal storage coefficients calculated by 2 S-TOOL and the coefficients calculated by Béjar-Pizarro et al. (2017)

To conclude, the results presented in this study differ slightly in magnitude from the values reported in previous papers, with most hydrogeological parameters fitting into the same order of magnitude, as shown in most of the figures above. Importantly, the use of data digitized from images, rather than original data, increases the data volume compared with that of the articles. This leads to variations in the least-squares fitting performed by the application and can explain the slight differences in the results. Nevertheless, it has been demonstrated that the application performs a good analysis of elastic and elastoplastic stress–strain curves, validating the proposed methodology for future studies in this field. This section highlights the differences between the characteristics of S_{ke} (obtained through the arithmetic mean) and the $S_{ke_weighted}$ coefficient (calculated by considering the total elastic points and the number of points in each elastic segment) for the analysed curves. Although these values are often similar, $S_{ke_weighted}$ has an advantage, as it assigns greater weight to larger loops, resulting in a more coherent result.

In summary, the developed open access application (2 S-TOOL) enhances and partly automates the analysis of the stress–strain curves and the inference of the characteristic coefficients. It has broad temporal and spatial applicability, allowing for the estimation of hydrogeological parameters for entire aquifer systems or specific layers (aquitards or aquifers). These parameters play crucial roles in water resource management. In the context of addressing the critical role of hydrogeological parameters in water resource management, this paper discusses the importance

of storage coefficients in assessing aquifer reserves and the volume of water that can be extracted from them (i.e., storage capacity and loss of an aquifer). With increasing water demand, land subsidence from aquifer exploitation could affect more than 484 million people by 2040 (Herrera-García et al. 2021), and quantifying and analysing land behaviour through stress–strain studies is vital. These studies can provide valuable insights for developing risk mitigation strategies in both urban and nonurban areas, potentially reducing infrastructure damage and saving significant costs.

At present, aquifer storage coefficients are usually determined through pumping tests at specific locations within the aquifer (Zhang et al. 2014). However, the proposed methodology, supported by the novel developed application, enables the estimation of the storage coefficient for the entire aquifer system if deformation data for the entire aquifer surface, obtained through InSAR or other measurement systems, are available. This approach would reduce the economic costs associated with well construction and pumping tests; it would also provide global hydrogeological parameters for the entire aquifer system (which can also be valuable for subsidence modelling purposes), and it would produce those advantages with a much more time-efficient methodology.

One important limitation of 2 S-TOOL is its inability to automatically separate the calculation coefficients for stress–strain curves that exhibit more than two mechanical behaviours (Fig. 5e–f). In cases where viscoelastoplastic behaviour is observed, it becomes necessary to divide the curve into segments or periods to accurately calculate the elastic and inelastic coefficients. However, this segmentation cannot be

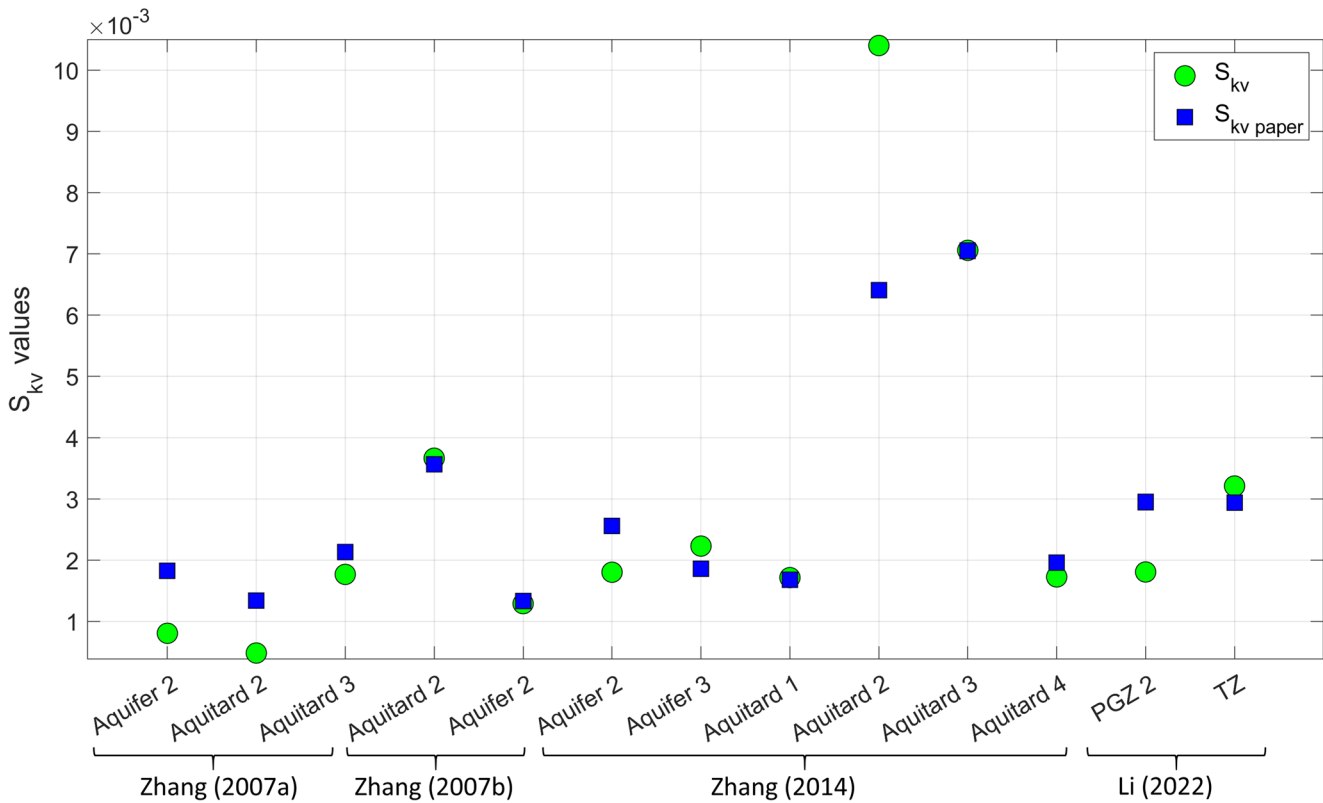


Fig. 12 Comparison between the inelastic skeletal storage coefficients calculated by 2 S-TOOL and the coefficients calculated by different authors

performed automatically with 2 S-TOOL. Instead, it requires the use of two separate input datasets corresponding to different periods, even though they refer to the same analysed layer. This limitation highlights the need for further development to enhance the ability of the application to process complex mechanical behaviours more efficiently. In addition, the process is not entirely automatic: the definition of certain input parameters still relies on user supervision and expert judgement, often after a phase of trial and error, to ensure reliable values for each case study. These aspects highlight the need for further development to improve both the capacity of the tool to process complex mechanical behaviours and its level of automation in parameter selection.

Additionally, in future research, an interesting way to evaluate whether the results obtained with this methodology correspond to the reality of aquifer systems would be to conduct a study comparing the hydrogeological parameters obtained through pumping tests with those obtained through the application.

Conclusions

As highlighted in this work, the deformation behaviour of an aquifer system due to the temporal evolution of its piezometric head, which is influenced by multiple

anthropogenic and/or natural factors, is a highly complex phenomenon. Therefore, the stress–strain relationships must be scientifically rigorous and as objective as possible. However, the absence of an objective methodology based on universally applicable criteria for all case studies and independent of subjective factors or interpretations by the analyst was the main motivation for developing the methodology proposed in this work and implementing it in open-source software.

The methodology proposed for the study of stress–strain curves, codified via MATLAB software, allows for the study of aquifers with elastic or elastoplastic behaviour. The results obtained during the validation process via information from several case studies are consistent, making the methodology applicable to all behaviour types. The application is based on a protocol that relies on criteria and parameters that allow for the adaptation of this methodology to a wide variety of case studies and geotechnical contexts. Similarly, it streamlines, automates, and reduces the subjectivity of the results. The proposed application not only reduces subjectivity but also provides results that are based on all the elastic loops identified and not only the most significant ones. Notably, the proper use of the application and user-controlled parameters requires prior knowledge of the magnitude of the subsidence phenomenon for the specific case study being analysed.

This methodology facilitates the automatic derivation of aquifer storage coefficients from stress–strain curves. This could support the implementation of measures for the rapid and effective management of groundwater resources. Moreover, the information derived from the use of this methodology and its computer application is highly useful for groundwater resource managers, as it enables the implementation of measures for the proper management of groundwater resources in a faster, more effective, and sustainable manner from a water resource perspective. Additionally, the parameters derived from stress–strain curves allow for the development of hydrogeological models that are applicable to such management.

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Author contributions María I. Navarro-Hernández: Methodology, Validation, Formal analysis, Data curation, writing – original draft, Writing – review & editing, Supervision. Sergio García-Pozo: Conceptualization, Methodology, Validation, Formal analysis, Data curation, Writing, Supervision. Javier Valdés-Abellán: Conceptualization, Methodology, Validation, Formal analysis, Data curation, Writing – review & editing, Supervision. Roberto Tomás: Conceptualization, Methodology, Writing – review & editing, Supervision, Funding acquisition. All authors reviewed the manuscript.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests María I. Navarro-Hernandez and reports financial support was provided by Horizon 2020 European Innovation Council Fast Track to Innovation (grant agreement No 1924). María I. Navarro-Hernandez and Roberto Tomás reports financial support was provided by European Space Agency and Ministry of Science and Technology of the People's Republic of China under ESA-MOST China DRAGON-5 (ref. 59339) and DRAGON-6 (ref. 95355) projects. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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